

Sharp Gaussian constants for row deletion in unit-row matrices

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

Analytic proof drafts give the Gaussian trimmed-second-moment limit for adversarial row retention in independent spherical matrices whenever $n \rightarrow \infty$ and $m/n \rightarrow \infty$, with explicit probability bounds. A projection argument proposes the same sharp universal asymptotic upper constant over all unit-row matrices, without an aspect-ratio requirement for the upper bound. These are analytic results; the accompanying finite verifier does not certify them.

1 Definitions and statements

Fix $0 < \theta < 1$, let $A \in \mathbb{R}^{m \times n}$ have unit Euclidean row norms, and set $k = \lfloor \theta m \rfloor$. Define

$$s_\theta(A) = \min_{|S|=k} \sigma_{\min}(A_S) = \min_{\|x\|_2=1} \left(\sum_{i=1}^k (|Ax|^2)_{(i)} \right)^{1/2},$$

where order is increasing, $\sigma_{\min}$ denotes the variational minimum over unit vectors (zero for a rank-deficient map), and empty sums are zero. The equality follows by interchanging the finite and compact minima.

Let $G \sim N(0, 1)$, $\mathbb{P}(|G| \leq a_\theta) = \theta$, and let φ be its density. Set

$$h_\theta = \mathbb{E}[G^2 \mathbf{1}_{|G| \leq a_\theta}] = \theta - 2a_\theta \varphi(a_\theta), \quad c_\theta = \sqrt{h_\theta}.$$

Theorem 1 (Random rows). *For independent uniform rows on S^{n-1} , along every sequence $n \rightarrow \infty$, $m/n \rightarrow \infty$, in probability*

$$\frac{n}{m} s_\theta(A)^2 \rightarrow h_\theta, \quad \frac{n}{m} \|A\|_2^2 \rightarrow 1, \quad \frac{s_\theta(A)^2}{\|A\|_2^2} \rightarrow h_\theta.$$

Theorem 2 (Universal constant). *Uniformly in all integers $m \geq 1$ and unit-row matrices $A \in \mathbb{R}^{m \times n}$,*

$$\sqrt{n/m} s_\theta(A) \leq c_\theta + o_{n \rightarrow \infty}(1).$$

No smaller constant has the eventual uniform upper-bound property. Along every $n \rightarrow \infty$, $m/n \rightarrow \infty$, the supremum over unit-row matrices of $\sqrt{n/m} s_\theta(A)$ converges to c_θ .

2 Trimming identities

For nonnegative $y_1, \dots, y_m$,

$$\frac{1}{m} \sum_{i=1}^k y_{(i)} = \max_{t \geq 0} \left\{ \frac{k}{m} t - \frac{1}{m} \sum_i (t - y_i)_+ \right\}.$$

The right side is concave piecewise linear, with slope k/m minus the empirical distribution function, and is maximized between the k th and $(k+1)$ st order statistics; use $t = 0$ for $k = 0$.

For a nonnegative random variable Y , define

$$H_\theta(Y) = \inf\{\mathbb{E}(Yw) : 0 \leq w \leq 1, \mathbb{E}w = \theta\}.$$

This is the integral of the lower quantiles through mass θ . For a continuous distribution it is $\mathbb{E}[Y\mathbf{1}_{Y \leq b_\theta}]$. Under any coupling of Y, Z ,

$$|H_\theta(Y) - H_\theta(Z)| \leq \mathbb{E}|Y - Z|.$$

Indeed use the same feasible selectors on the joint space. Allowing selectors to depend on both variables does not change either infimum: conditional expectation given one variable preserves the constraints.

3 Pointwise concentration

Let u be uniform on S^{n-1} , $n \geq 2$, $Y_n = nu_1^2$, and $h_{\theta,n} = H_\theta(Y_n)$. Its law is continuous and its mean is one. Put $L = 2/(1 - \theta)$. For independent copies Y_i , $T_m = m^{-1} \sum_{i=1}^k Y_{(i)}$, and $0 < \epsilon \leq (1 - \theta)/2$, we claim

$$\mathbb{P}(|T_m - h_{\theta,n}| > 2L\epsilon + L/m) \leq 5e^{-2m\epsilon^2}.$$

Let b be the θ -quantile. Since the mean is one, $b \leq 1/(1 - \theta) \leq L$. Hoeffding's bounded-variable inequality gives, outside total probability $4e^{-2m\epsilon^2}$,

$$\left| m^{-1} \sum_i \mathbf{1}_{Y_i \leq b} - \theta \right| \leq \epsilon, \quad \left| m^{-1} \sum_i Y_i \mathbf{1}_{Y_i \leq b} - h_{\theta,n} \right| \leq b\epsilon.$$

Markov's inequality gives $\mathbb{P}(Y_n \leq L) \geq (1 + \theta)/2$. The chance fewer than k samples are at most L is at most $e^{-m(1-\theta)^2/2} \leq e^{-2m\epsilon^2}$. On the complementary events, replacing the sum below b by the k smallest terms changes at most $m\epsilon + 1$ terms, each at most L . This proves the claim. For $k = 0$, use $h_{\theta,n} \leq \theta < 1/m$.

The bounded-variable inequality used here follows from $\mathbb{E}e^{\lambda(X - \mathbb{E}X)} \leq e^{\lambda^2 b^2/8}$ for $0 \leq X \leq b$, independence, and optimization of the Chernoff parameter. The scalar exponential bound follows by convexity on an interval and a one-variable maximization.

4 Covariance and uniform directions

The moments satisfy

$$\mathbb{E}Y_n^r = \frac{n^r(2r-1)!!}{n(n+2)\cdots(n+2r-2)} \leq 2^r r!,$$

so $\mathbb{E}|Y_n - 1|^r \leq 4^r r!$ and, for $|\lambda| \leq 1/8$, $\mathbb{E}e^{\lambda(Y_n - 1)} \leq e^{32\lambda^2}$. The mean-zero linear term vanishes; summing the remaining geometric series gives this bound. Chernoff's inequality then yields, for $0 < u \leq 1$,

$$\mathbb{P}\left(\left|m^{-1} \sum_i Y_i - 1\right| > u\right) \leq 2e^{-mu^2/128}.$$

A $1/4$ -net of the real unit sphere has at most 9^n points; a symmetric matrix's norm is at most twice its maximum absolute quadratic form on that net. For $C = (n/m)A^T A$ this gives

$$\mathbb{P}(\|C - I\|_2 > t) \leq 29^n e^{-mt^2/512} \quad (0 < t \leq 1).$$

The net cardinality follows by disjoint-ball volume comparison; the quadratic-form estimate follows by approximating a maximizing unit vector.

Let $T(x) = (n/m) \min_{|S|=k} \|A_S x\|_2^2$. On $\|C - I\|_2 \leq t$, every submatrix quadratic form has norm at most $1 + t$, hence

$$|T(x) - T(y)| \leq 2(1+t) \|x - y\|_2$$

for unit x, y . A δ -net has at most $(1 + 2/\delta)^n$ points. Pointwise concentration, a union bound, and this Lipschitz estimate give

$$\sup_{\|x\|_2=1} |T(x) - h_{\theta,n}| \leq D := 2L\epsilon + L/m + 2(1+t)\delta$$

except with probability

$$29^n e^{-mt^2/512} + 5(1 + 2/\delta)^n e^{-2m\epsilon^2},$$

where $0 < t < 1$, $0 < \delta < 1$, $0 < \epsilon \leq (1 - \theta)/2$.

5 Gaussian limit and explicit random error

Couple u with an independent radius $R^2 \sim \chi_n^2$. Then Ru is standard Gaussian, so $G_1^2 = (R^2/n)Y_n$. Independence and the chi-square variance give

$$\mathbb{E}|Y_n - G_1^2| = \mathbb{E}|1 - R^2/n| \leq \sqrt{2/n}, \quad |h_{\theta,n} - h_\theta| \leq \sqrt{2/n}.$$

With the preceding exceptional probability,

$$\begin{aligned} \left| \frac{n}{m} s_\theta(A)^2 - h_\theta \right| &\leq D + \sqrt{2/n}, \\ \left| \frac{s_\theta(A)^2}{\|A\|_2^2} - h_\theta \right| &\leq \frac{D + \sqrt{2/n} + t}{1 - t}. \end{aligned}$$

Here $h_\theta \leq 1$. If $Q = m/n \rightarrow \infty$, eventually choose $t = \epsilon = \delta = 32\sqrt{\log Q/Q}$. Parameter restrictions eventually hold; both failure probabilities vanish and the error is

$$O_\theta \left(\sqrt{\frac{n}{m} \log \frac{m}{n}} + n^{-1/2} + m^{-1} \right).$$

No additional condition on how quickly Q diverges is needed. This proves the random theorem.

6 Deterministic projection argument

For arbitrary unit-row A , choose $1 \leq r < n$, set $d = n - r$, and remove the r largest covariance eigenvector directions. In orthonormal coordinates on the remaining subspace, call the projected matrix $B \in \mathbb{R}^{m \times d}$. Since $\text{tr}(A^T A) = m$,

$$\|B\|_2^2 \leq m/(r+1), \quad \|b_i\|_2 \leq 1, \quad \text{tr}(B^T B) \leq m.$$

Let $g \sim N(0, I_d)$ and $Y_i = (b_i^T g)^2$; these variables may be correlated. Define

$$\Psi_t(g) = \frac{k}{m}t - \frac{1}{m} \sum_i (t - Y_i)_+.$$

Each scalar Gaussian $b_i^T g$ has variance at most one, so

$$\mathbb{E}\Psi_t(g) \leq \theta t - \mathbb{E}(t - G^2)_+ \leq h_\theta.$$

The final inequality is the population trimming formula, maximized at $t = a_\theta^2$.

Gaussian Poincaré gives $\text{Var}(f(g)) \leq \mathbb{E} \|\nabla f(g)\|_2^2$. A proof expands smooth functions in tensor Hermite polynomials: variance sums squared nonconstant coefficients, while the gradient norm weights them by total degree, at least one. Smooth approximation extends the inequality to Lipschitz functions. Since the scalar derivative of $(t - z^2)_+$ is bounded by $2\sqrt{t}$,

$$\text{Var}(\Psi_t(g)) \leq \frac{\|B\|_2^2}{m^2}(4tm) \leq \frac{4t}{r+1}.$$

This handles arbitrary row correlations without asserting independence.

7 Explicit uniform bound and sharpness

For $0 < \delta < 1$, a grid of $[0, L]$ with spacing at most δ has at most $L/\delta + 2$ points. Chebyshev's inequality and the last variance bound show that all grid values obey $\Psi_t \leq h_\theta + \delta$, except with probability at most

$$\frac{4L(L/\delta + 2)}{(r+1)\delta^2}.$$

Since $t \mapsto \Psi_t$ is 1-Lipschitz, its supremum on $[0, L]$ is at most $h_\theta + 2\delta$ on that event's complement.

Also $Z = m^{-1} \|Bg\|_2^2$ has mean at most one and variance $2 \text{tr}((B^T B)^2)/m^2 \leq 2/(r+1)$. Thus $Z \leq 2$ except with probability $2/(r+1)$. On that event, the k th order statistic is at most L , because at least $m - k + 1 \geq (1 - \theta)m$ terms are as large. Therefore the full trimming maximum is attained in $[0, L]$; the case $k = 0$ is immediate.

Finally, $\mathbb{P}(\|g\|_2^2 < (1 - \delta)d) \leq 2/(d\delta^2)$. Consequently, if

$$\frac{2}{r+1} + \frac{4L(L/\delta + 2)}{(r+1)\delta^2} + \frac{2}{(n-r)\delta^2} < 1,$$

there exists a realization satisfying all the bounds. Normalize that g to a unit direction in the retained subspace. The trimming identity yields

$$\boxed{\frac{n}{m} s_\theta(A)^2 \leq \frac{n}{(n-r)(1-\delta)} (h_\theta + 2\delta).}$$

All constants are independent of m, A . Taking $r = \lfloor n^{2/3} \rfloor$, $\delta = n^{-1/6}$ makes the condition hold for all sufficiently large n for fixed θ ; its largest term is $O_\theta(n^{-1/6})$. The result is the uniform bound $(n/m) s_\theta(A)^2 \leq h_\theta + O_\theta(n^{-1/6})$.

The random theorem supplies deterministic realizations arbitrarily close to h_θ along every high-aspect-ratio sequence. Thus no smaller constant can satisfy the eventual uniform requirement, and the normalized supremum converges to c_θ . This proves the second theorem's two quantifier claims without interchanging an unjustified random minimum and expectation.

8 Sources and verification limits

Recovered target descriptions: *Sharp row-deletion constant for spherical random matrices*, provisionally IE-21, and *Optimal uniform row-deletion constant*, provisionally IE-22, in OpenProblemsInNLA.¹ Earlier conversational summaries assigned different identifiers to these topics; the present labels are local identifiers only until the repository mapping is checked.

The live source statements could not be retrieved during reconstruction. The precise normalization, retention convention, singular-value definition, and quantifiers in Section 1 are essential. The probability and existence arguments need independent mathematical review; finite tests of the other notes do not certify them.

¹<https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/linear-systems-and-elimination>.